

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTECPE603B - BIG DATA ANALYTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 Which of the following is not a type of digital data?

- A) Structured data
- B) Unstructured data
- C) Semi-structured data
- D) Discrete data

Hadoop was originally developed by

- A) IBM
- B) Google
- C) Apache Software Foundation
- D) Yahoo!

3 What is the main advantage of using Hadoop for Big Data processing?

- A) It requires expensive hardware
- B) It can only process structured data
- C) It allows distributed storage and parallel processing
- D) It runs only on Windows OS

4 Which of the following tools is used for importing data from relational databases into HDFS?

- A) Flume
- B) Hive
- C) Sqoop
- D) Pig

What is the main purpose of Hadoop Archives (HAR files)?

- A) To compress files for transmission
- B) To allow real-time analytics
- C) To store large numbers of small files efficiently
- D) To reduce Namenode memory usage

6 In Hadoop, which of the following is used for streaming log data into HDFS in real time?

- A) Sqoop
- B) Pig
- C) Flume
- D) Hive

7 In a MapReduce job, what is the main purpose of the shuffle and sort phase?

- A) To combine multiple jobs
- B) To distribute data across clusters
- C) To transfer map output to reducers and sort it by key
- D) To delete temporary files

- 8 **What happens if a MapReduce task fails during execution?**
 A) The entire job is aborted immediately
 B) It is automatically re-executed on another node
 C) Hadoop ignores the failed task
 D) The user is prompted to restart the task manually
- 9 **In Hive, what is the role of the Metastore?**
 A) It executes Hive queries
 B) It stores the actual data files
 C) It manages metadata about tables, partitions, and schemas
 D) It loads data into HDFS
- 10 **Which of the following is the scripting language used in Apache Pig?**
 A) SQL
 B) Pig Latin
 C) HiveQL
 D) Grunt
- 11 **In Apache Kafka, which component is responsible for receiving data from producers and storing it?**
 A) Zookeeper
 B) Broker
 C) Consumer
 D) Topic
- 12 **In Apache Spark, what is the function of the Driver Program?**
 A) It stores all the data
 B) It manages cluster resources
 C) It coordinates tasks and maintains the SparkContext
 D) It serves as a backup node
- Q2. Solve the following.
 A) Explain the different types of digital data. How does the variety of data influence Big Data analytics? 6
 B) Describe the history and evolution of Hadoop. How has Apache Hadoop transformed big data processing? 6
- Q3. Solve the following.
 A) Explain the design and key concepts of HDFS. What are its core components, and how does it ensure fault tolerance and high availability? 6
 B) Describe the data ingest methods in Hadoop using Flume and Sqoop. Compare their purposes and use cases. 6
- Q4. Solve Any Two of the following.
 A) What are the common types of failures in a MapReduce job, and how does Hadoop handle them? 6
 B) What are the different MapReduce data types and formats supported in Hadoop? Explain with examples. 6
 C) What are the features of a Hadoop cluster? Describe the components and their roles in executing MapReduce jobs. 6
- Q5. Solve Any Two of the following.
 A) Differentiate between the local mode and MapReduce mode of Pig execution. In which scenarios would each be used? 6
 B) Write a short note on Hive Shell and HiveQL. How does Hive differ from traditional RDBMS? 6
 C) What is Big SQL? Describe its significance and how it enables SQL queries on Hadoop data. 6

Q6. Solve Any Two of the following.

- A) Explain the architecture of Apache Kafka with a neat diagram. Describe the roles of its key components: Producer, Broker, Consumer, Topic, and Zookeeper. 6
- B) Describe the architecture of Apache Spark. Explain the roles of the Driver Program, Cluster Manager, and Executors. 6
- C) What is Kerberos authentication? Explain its key components and how they work together to provide secure access. 6

025333211

*** **

025333211

025333211

025333211

025333211

025333211

025333211

025333211

025333211